

A Look at South Dakota Flood Recovery – Black Hills (Rapid City, SD) Flood 1972

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INTRODUCTION

Recovery from disaster requires significant planning, coordination, and resource. Floods are the most frequent natural disaster in the United States and the entire population lives in a flood zone. Using the 1972 Black Hills Flood as a site study, modern emergency recovery process data is reported via historic patterns. Destruction models are described and displayed using GPS and current spatial mapping software (ESRI ArcMap).

This document serves to explore an historic disaster (flood) in the Black Hills Region of South Dakota, U.S. and its recovery efforts in order to display present day emergency response techniques and efforts.

METHODS

Data Used

For purpose of discussion, the data taken in this study is from an historic event. Using the 1972 Black Hills (Rapid City, South Dakota) Flood and nearby sites as the sample. No direct field work was collected, this research is based on data published and historic data gathered. By using coordinate points for effected areas in Rapid City, display using Google Earth and ArcMap presentation, and archived visual aerial photography/imagery we can understand how a disaster example provides modern recovery procedure improvements.

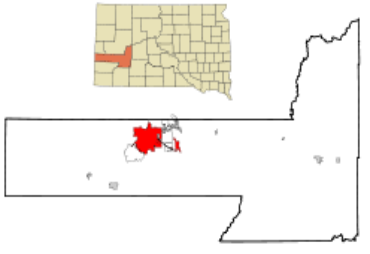


Black Hills Flood damage immediately after event. Four Presidents' Motel was torn down after flooding. Keystone, South Dakota.



Heavy structural highway damage. South Dakota National Guard maintenance and rebuild Efforts after 1972 Black Hills (Rapid City, South Dakota) Flood.

Data for this study is simply taken from personal and internet research. Using Google Earth images and coordinate points before, during, and presently, we're able to show site pre and post event. The above images show damage immediately after the flood in Rapid City and display quick efforts at recovery in road rehabilitation. This project will show small town site building damage numbers pre and post flood event. Between June 9th and 10th 1972, nearly 15 inches of rain fell in and around Rapid City, South Dakota at rapid enough pace to cause water levels to rise 3+ feet in as little as 15 minutes. Ensuing damage destroyed 1,335 homes and killed 238 people; 8 of which were in the town of Keystone. Recovery cost nearly \$1 billion.



Rapid City, South Dakota "Black Hills." Flood Area.

Beginning right after a disaster, recovery looks to return a community to normal in as little time as possible. The data and mapping presented in this paper serve to demonstrate spatial changes in a small flood site in Keystone, South Dakota.

ANALYSIS

While satellite imagery was not available in depth during 1972, this study will operate under the archived store and resident points compared with modern coordinate building points through Google Earth mapping. Taking observed building points just before, then after, and then currently help display recovery efforts and score give for a site. Using residential building points on and near Blair Street in Keystone, spatial mapping with mark negative and/or negative block reproduction based on imagery to point analysis.

RESULTS

Little data, let alone spatial imagery is available for 1972 South Dakota building placement, but using site placement data from South Dakota state records and disaster measures help create human and building displacement after the 1972 flood. The figures below (a – c) provide a visual for the building placement, loss, and recovery at different dates via Google Earth imagery.



Figure a. represents pre-flood standing building placement. Figure b. serves to show building change just after the flood had subsided. Only one building remains on Blair Street. Finally, figure c. presents current building conditions on the Keystone, South Dakota point.

Currently nine buildings stand where three existed pre-event and one just post-event. Where more measures are necessary to see the progression, we are still able to see the gradual build post-disaster.

DISCUSSION

The South Dakota Department of Health and Response OEMs team and train directly with FEMA and national level disaster recovery units. While this study exemplifies a much older disaster, it is meant to represent the progression of modern local disaster recovery measures and techniques. As U.S. population expands, the necessity for natural disaster preparedness to recovery rises. Meshing standards applied from state to federal levels enhances human recovery efforts. In order to return to normalcy post-flood, cleanup and human-assistance must be addressed efficiently. Better data acquisition and presentation allow for continual growth in this concern.

CONCLUSIONS

Using sites in Pennington County South Dakota, a classic disaster example (1972 Black Hills Flood, June 9-10) and spatial resources such as Google Earth, classic and current data can be merged and therefore presented to display the advancement of recovery efforts for disasters in the western South Dakota area. With tools powerful analysis tools such as ESRI's ArcMap in conjunction with available spatial data (U.S. Census Bureau), the above map results illustrate repeatable emergency recovery protocol specific to the study area. By taking block site data and satellite imagery pre-flood, just after flood, and near current conditions and merging within mapping software, data is integrated and easily updated for modern flood recovery operations. This site data serves a recovery example for state and local emergency teams to build upon.